

	
3(a)	Electric current is the rate of flow of charge.
3(b)	All the charge carriers within the volume of conductor of length l will pass the cross-sectional area A during the time t. I = rate of flow of charge $= (\text{rate of flow of charge carriers}) \times (\text{charge on each charge carrier})$ $= \left[(\text{number of charge carriers per unit volume}) \times (\text{volume flow rate}) \right] \times q$ $= n \left(\frac{\text{volume}}{\text{time}} \right) q$ $= n \left(\frac{Al}{t} \right) q$ $= nA \left(\frac{l}{t} \right) q$ $= nAvq$

Qn	Solution
3(c)(i)	$I = nAvq$ $2.00 = \frac{1.45 \times 10^{23}}{Al} Av (1.6 \times 10^{-19})$ $2.00 = \frac{1.45 \times 10^{23}}{A(3.00)} Av (1.6 \times 10^{-19})$ $\therefore v = 2.59 \times 10^{-4} \text{ m s}^{-1}$
3(c)(ii)	The high mean speed of a conduction electron is due to its random motion inside the tungsten wire, which is present even if the potential difference across the wire is zero. Drift velocity is the velocity attained by the conduction electrons due to the electric field inside the wire when a potential difference is applied across the ends of the wire. It is the average velocity of all the electrons in one direction along the wire. The conduction electrons collide with the metal lattice ions back and forth as they drift from the lower potential end of the wire to the higher potential end of the wire. This leads to lower drift velocity compared to the mean speed of the conduction electron.
4(a)	Magnetic flux density (B) is defined as the force acting per unit current per unit